



NATIONAL TEXTILE UNIVERSITY

Department of Computer Science

Lab # 2: Programming Fundamentals (COC-1071)

Basic Information			
Registration#		Name	
Total Marks		Marks Obtained	
Tools:	Dev-C++ 5.4.1		
Objectives	1. Variables 2. If statements	3. Relational and arithmetic operators	
Note	Solve the following problems using the concepts we have covered so far		
Activity 1: Variables and Data Types			

1. Write a C++ program to calculate the area of a rectangle given its length and width.
2. Create a program that converts temperature from Fahrenheit to Celsius and vice versa.
3. Write a program that calculates the average of three numbers entered by the user.
4. Develop a program that swaps the values of two integer variables without using a third variable.
5. Write a C++ program to compute the volume of a sphere using the radius provided by the user.
6. Create a program that converts a given number of days into years, months, and days.
7. Write a program that takes a user's age as input and then prints the age in months and days.
8. Create a program that converts a given number of minutes into hours and minutes.
9. Write a program that calculates the sum and product of two numbers entered by the user.
10. Develop a program that computes the area of a triangle using the lengths of its three sides (Heron's Formula).

Activity 2: Operators and Expressions

11. Write a C++ program that calculates the area of a triangle when the lengths of its base and height are given.
12. Create a program that checks whether a given number is even or odd and prints the result.
13. Write a program that checks if a given year is a leap year or not.
14. Develop a program that calculates the simple interest for a loan or deposit.

15. Write a C++ program that converts a decimal number to binary and prints the binary representation.
16. Create a program that checks if a given number is a prime number or not.
17. Write a program that calculates the factorial of a non-negative integer.
18. Develop a program that calculates the power of a number (a^b).
19. Write a C++ program that calculates the roots of a quadratic equation ($ax^2 + bx + c = 0$).
20. Create a program that takes an integer input and prints whether it is positive, negative, or zero.
21. Write a C++ program that takes a user's age as input and prints "You are a minor" if the age is less than 18, and "You are an adult" if the age is 18 or older.